

12.268

RAVULA SHASHANK REDDY - EE25BTECH11047

October 12, 2025

Question

A 3×3 matrix has elements such that its trace is 11 and its determinant is 36. The eigenvalues of the matrix are all known to be positive integers. Find the largest eigenvalue of the matrix.

Solution

Let the eigenvalues of the 3×3 matrix $\mathbf{A}$ be $\lambda_1, \lambda_2, \lambda_3$.

We are given:

$$\text{trace}(\mathbf{A}) = \lambda_1 + \lambda_2 + \lambda_3 = 11, \quad (1)$$

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \lambda_3 = 36 \quad (2)$$

$$\lambda_1, \lambda_2, \lambda_3 > 0 \quad (3)$$

Check the possible factor triplets of 36 and their sums:

$$1 + 4 + 9 = 14 \neq 11 \quad (4)$$

$$2 + 2 + 9 = 13 \neq 11 \quad (5)$$

$$2 + 3 + 6 = 11 \quad (6)$$

$$3 + 3 + 4 = 10 \neq 11 \quad (7)$$

Hence, the eigenvalues are:

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}. \quad (8)$$

Thus, the largest eigenvalue is:

$$\lambda_{\max} = 6.$$

```
#include <stdio.h>

int main() {
    int trace = 11;
    int det = 36;
    int i, j, k;

    printf(The positive integer eigenvalues are:\n);

    for(i = 1; i <= trace; i++) { // possible first eigenvalue
        for(j = 1; j <= trace - i; j++) { // second eigenvalue
            k = trace - i - j; // third eigenvalue from trace
            if(k > 0 && i * j * k == det) { // check product =
                determinant
                printf(Eigenvalues: %d, %d, %d\n, i, j, k);
            }
        }
    }
}
```

```
        int max = i;
        if(j > max) max = j;
        if(k > max) max = k;
        printf(Largest eigenvalue: %d\n, max);
    }
}

return 0;
}
```

Python Code

```
import numpy as np

trace = 11 # sum of eigenvalues
det = 36 # product of eigenvalues

valid_eigenvalues = []

# Loop over all possible positive integers for first two
eigenvalues
for i in range(1, trace + 1):
    for j in range(1, trace - i + 1):
        k = trace - i - j
        if k > 0 and i * j * k == det:
            eigenvalues = np.array([i, j, k])
            valid_eigenvalues.append(eigenvalues)
            print(Eigenvalues:, eigenvalues)
            print(Largest eigenvalue:, np.max(eigenvalues))
if valid_eigenvalues:
    largest_overall = max(np.max(triplet) for triplet in
```

```
import ctypes

# Load the shared library
lib = ctypes.CDLL('./eigen.so')

# Define the function argument and return types
lib.find_largest_eigen.argtypes = [ctypes.c_int, ctypes.c_int]
lib.find_largest_eigen.restype = ctypes.c_int

# Inputs
trace = 11
det = 36

# Call the C function
largest = lib.find_largest_eigen(trace, det)

print(fLargest eigenvalue for trace={trace}, det={det} is: {
    largest})
```